

Biometric Fraud Pattern Analysis Report

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1. Executive Summary

Analysis period: 2024-01-01 to 2024-01-30 (30 days).
Authentication events analysed: 7,327. Fraud events in dataset: 414.
Alerts raised: 428 (about 14 per day).
Detected 408 of 414 fraud events (recall 98.6%) with 20 false positives (precision 95.3%).
39 duplicate impossible-travel alerts were correlated into existing incidents instead of raising new alerts.

The weakest-detected pattern was 'low_score_bypass' (96.4% recall). The 6 missed events should be reviewed to improve coverage.

2. Detection Performance

Detection combines rule-based checks (low match score, rapid retries, off-hours access, per-user score deviation, speed-based impossible travel) with an Isolation Forest anomaly model. Rules provide explainable reasons for each alert; the model catches unusual combinations the rules do not encode.

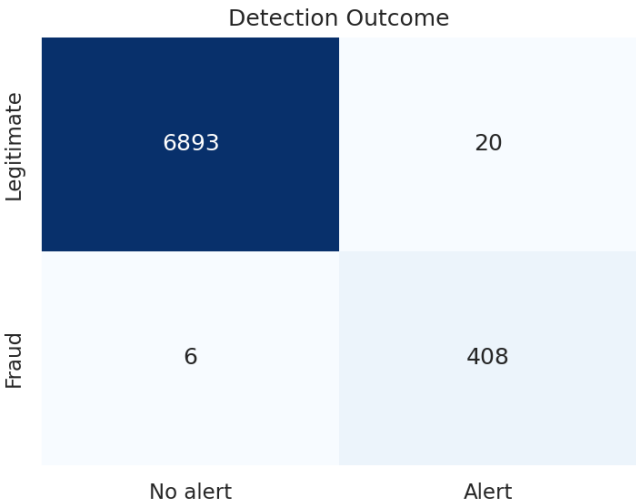


Figure 1: Detection outcome

3. Fraud Pattern Breakdown

Pattern	Events	Detected	Recall
geo_jump	114	114	100.0%
low_score_bypass	110	106	96.4%
off_hours	106	106	100.0%
rapid_retry	84	82	97.6%

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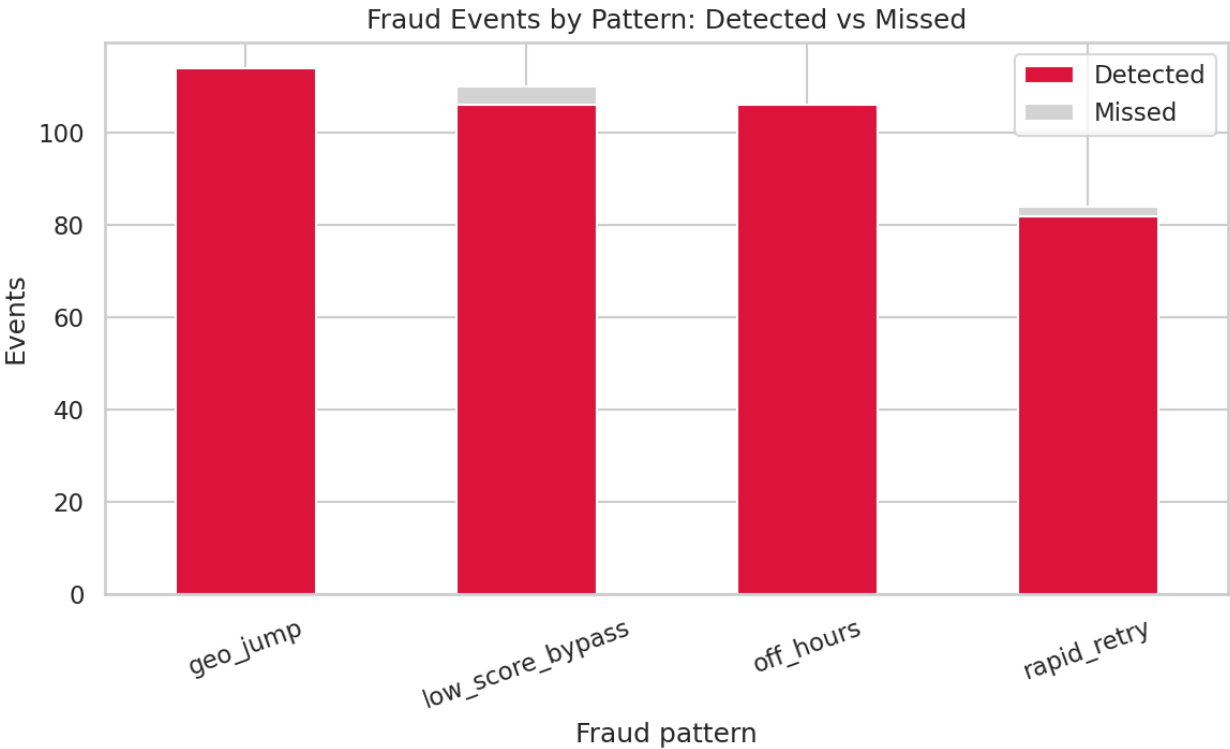


Figure 2: Fraud events by pattern

4. Match Score Analysis

Median match score: legitimate 0.82, fraud 0.71. 38.4% of fraud events scored below the 0.55 rule threshold. Fraud that passes with a normal-looking score (e.g. off-hours or geo-jump sessions) is caught by behavioural and travel signals, not by score alone.

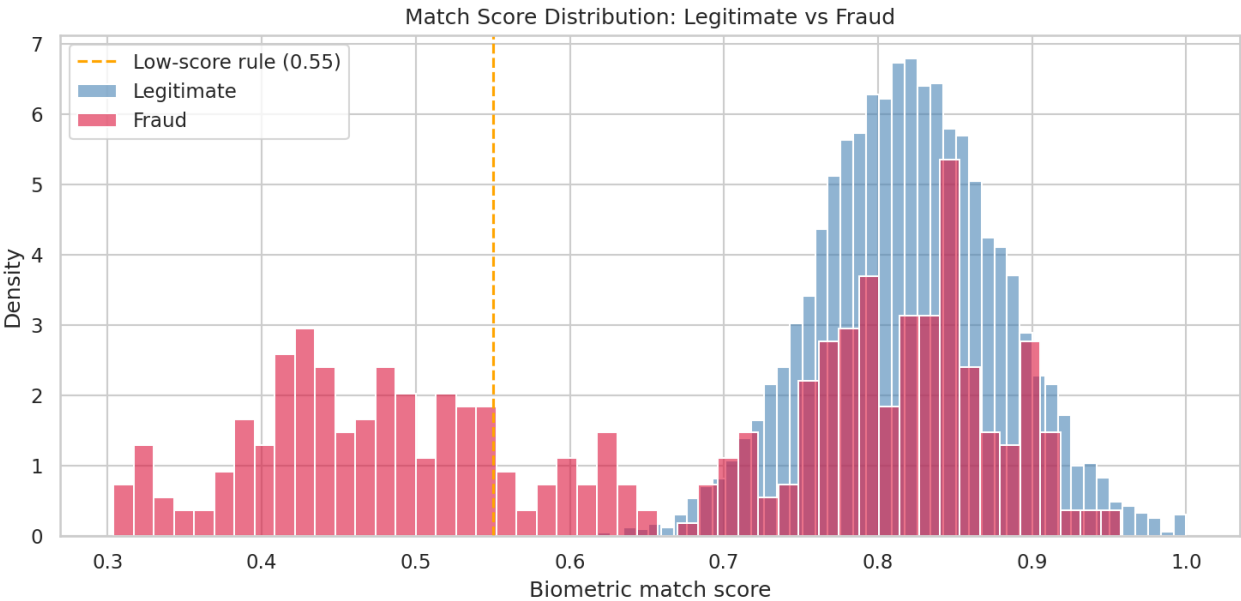


Figure 3: Match score distribution

5. Temporal Analysis

Alert volume averaged 14.3 per day, peaking at 22 alerts on 2024-01-01.

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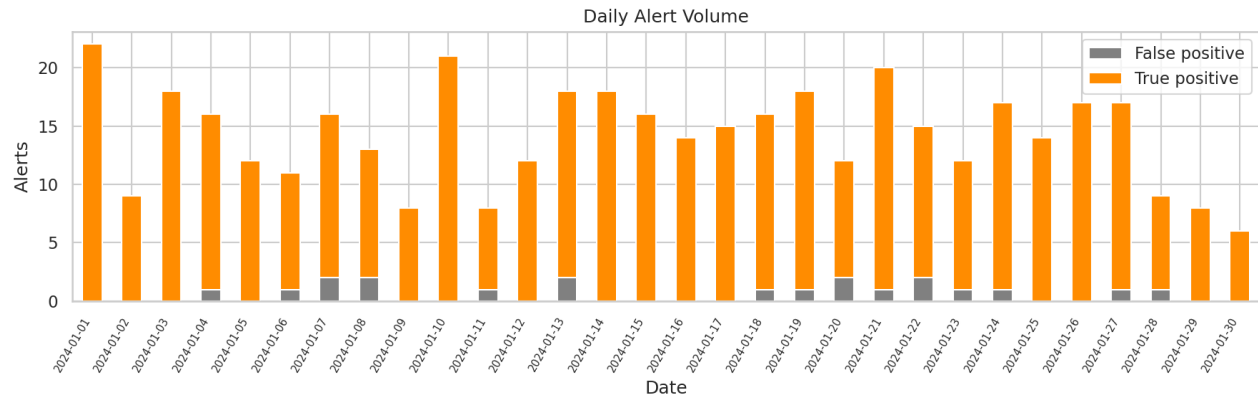


Figure 4: Daily alert volume

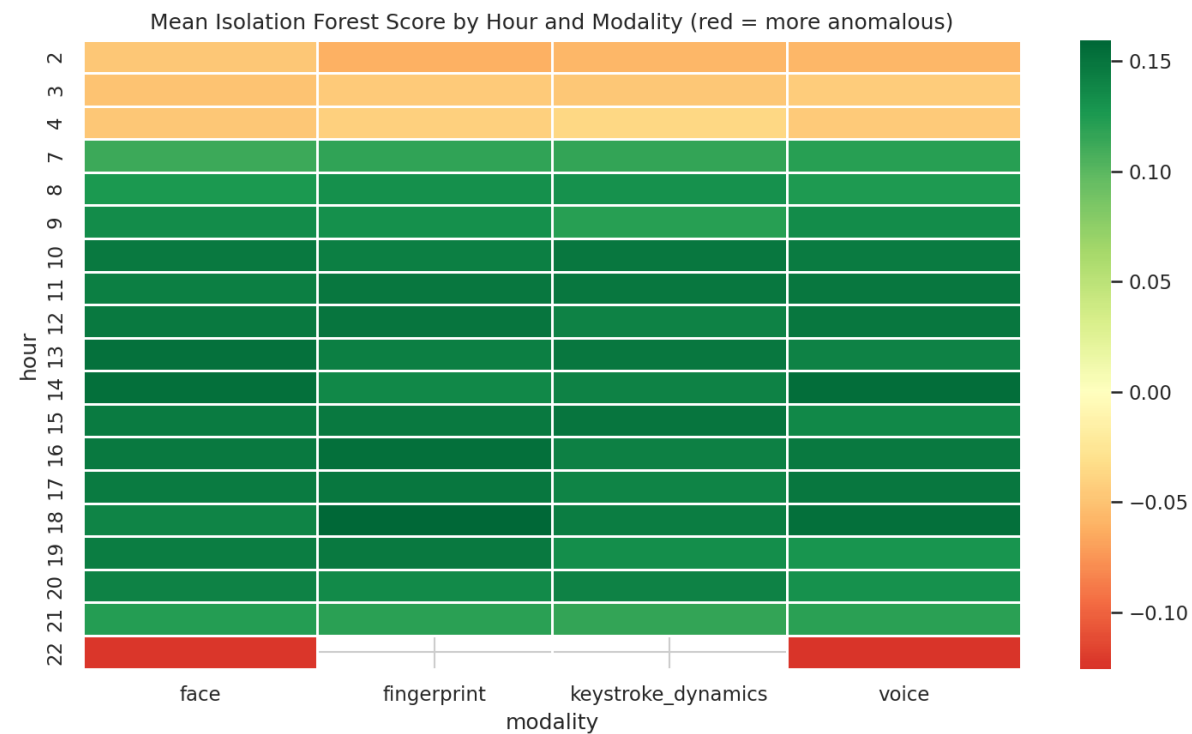


Figure 5: Mean anomaly score by hour and modality (red = more anomalous)

6. Accounts Recommended for Investigation

User	Alerts	Mean score	Locations seen
USR-0085	8	0.70	Lagos, London, Singapore
USR-0171	7	0.66	New York, Singapore
USR-0021	6	0.54	Lagos, Manchester
USR-0077	6	0.69	Berlin, Manchester, Singapore
USR-0106	6	0.61	Lagos, Manchester
USR-0097	6	0.72	Lagos, New York, Singapore
USR-0092	5	0.53	London, New York, Singapore
USR-0122	5	0.70	Berlin, Manchester, New York
USR-0066	5	0.59	Lagos, London
USR-0027	5	0.70	Lagos, London

7. Recommendations

1. Require step-up verification for match scores below 0.65. This would affect 0.3% of legitimate logins and 47.1% of fraudulent ones.
2. Enforce step-up authentication or temporary lockout when attempt count exceeds 5.
3. Keep speed-based impossible-travel detection (>900 km/h) with return-leg correlation; add a known-VPN/proxy allowlist before production use.
4. Treat off-hours access (23:00-06:00) as a corroborating signal and require MFA rather than blocking outright.
5. Retrain the Isolation Forest monthly and tune its sensitivity against analyst feedback on closed alerts.

8. Methodology and Limitations

All data is synthetic, generated with known fraud patterns so detection quality can be measured. Real traffic is noisier: legitimate users travel, use VPNs and log in at unusual hours. The anomaly model's sensitivity (6%) is close to the dataset's true fraud rate, which a production system would not know in advance. Tuning history is recorded in `reports/tuning_log.md`. In this dataset, hours 02-04 and 22 contain only fraud events, so the red rows in Figure 5 reflect how the data was generated, not a real-world finding.